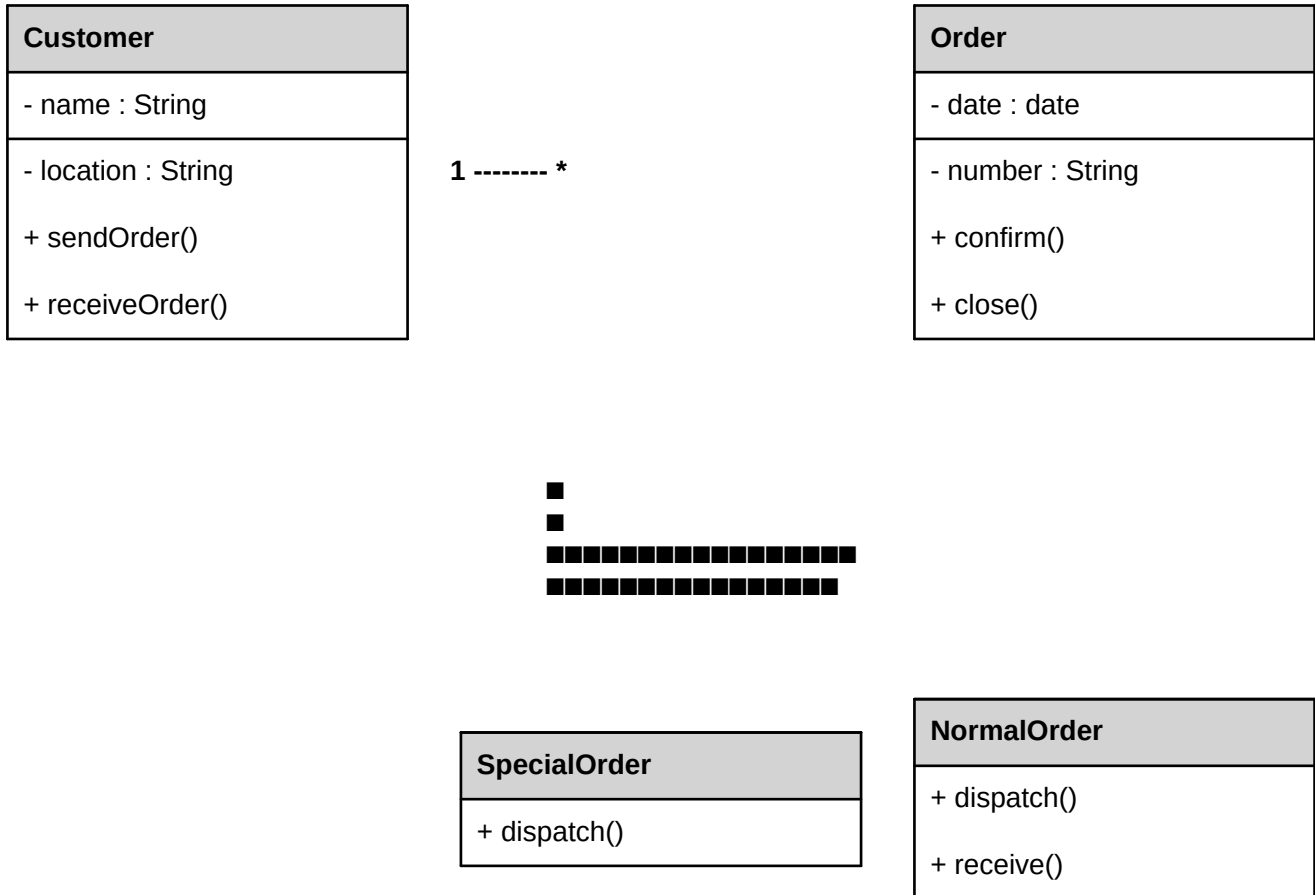


UML Class Diagram and Algorithm

Java Order Management Program

1. UML Class Diagram



Relationships: Customer is associated with one or more Orders. SpecialOrder and NormalOrder inherit from Order.

2. Algorithm

Step 1: Start.

Step 2: Create the Customer class with name and location attributes.

Step 3: Define sendOrder() and receiveOrder() methods in Customer.

Step 4: Create the Order class with date and number attributes.

Step 5: Define confirm() and close() methods in Order.

Step 6: Create SpecialOrder as a subclass of Order and define dispatch().

Step 7: Create NormalOrder as a subclass of Order and define dispatch() and receive().

Step 8: In main(), create a Customer object.

Step 9: Create one SpecialOrder object and one NormalOrder object.

Step 10: Display customer details.

Step 11: For the special order, call sendOrder(), confirm(), dispatch(), close(), and receiveOrder().

Step 12: For the normal order, call sendOrder(), confirm(), dispatch(), receive(), close(), and receiveOrder().

Step 13: Stop.

3. OOP Concepts Used

- Class and Object
- Inheritance — SpecialOrder and NormalOrder extend Order
- Constructors
- Methods
- Association — Customer interacts with Order objects